Test 1: Sept 24

HW 2: due Sept 17

Green's Functions

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$$

$$u(x) = \int_U G(x,y) f(y)\,dy - \int_{\partial U} \frac{\partial G}{\partial \nu_y}(x,y)\, g(y)\, dS(y)$$

$$\Phi(x-y) = c_n |x-y|^{2-n}$$

$$G(x,y) = \Phi(x-y) - \phi^x(y)$$

$$\begin{cases} -\Delta_y \phi^x(y) = 0, & y \in U \\ \phi^x(y) = \Phi(x-y), & y \in \partial U \end{cases}$$

If $f = 0$,

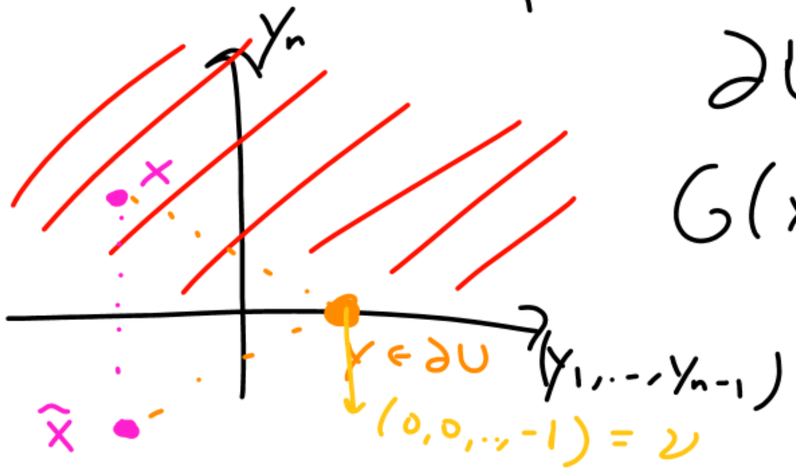
$$u(x) = -\int_{\partial U} \boxed{\frac{\partial G}{\partial \nu_y}(x,y)}\, g(y)\, dS(y)$$

→ Harmonic fcns can be written as integral involving boundary condition.

If $U = \mathbb{R}^n_+ = \{x \in \mathbb{R}^n : x_n > 0\}$,

$\partial U = \{x_n = 0\}$

$G(x,y) = \Phi(x-y) - \phi^x(y)$



$\tilde{x}_n = (x_1, x_2, \dots, x_{n-1}, -x_n)$

Let $\phi^x(y) = \Phi(\tilde{x} - y)$,

$G(x,y) = \Phi(x-y) - \Phi(\tilde{x}-y)$

$|x-y| = |\tilde{x}-y|$

$\Rightarrow \quad c_n |x-y|^{2-n} = c_n |\tilde{x}-y|^{2-n}$

$$\frac{\partial G}{\partial \nu_y}(x,y)$$

$$= -D_y G(x,y) \cdot (0,0,\dots,0,-1)$$

$$= -\frac{\partial}{\partial y_n} G(x,y)$$

$$= -\frac{\partial}{\partial y_n}\left[c_n |x-y|^{2-n} - c_n |\tilde{x}-y|^{2-n}\right]$$

$$= -c_n\left[(2-n)|x-y|^{1-n}\frac{\partial}{\partial y_n}|x-y| - (2-n)|\tilde{x}-y|^{1-n}\frac{\partial}{\partial y_n}|\tilde{x}-y|\right]$$

$$\frac{\partial}{\partial y_n}|x-y| = \frac{\partial}{\partial y_n}\sqrt{(x_1-y_1)^2+\cdots+(x_n-y_n)^2}$$

$$= \frac{1}{2}\left((x_1-y_1)^2+\cdots+(x_n-y_n)^2\right)^{-1/2}\cdot\left(-2(x_n-y_n)\right)$$

$$= \frac{-(x_n-y_n)}{|x-y|}, \quad \text{and}$$

$$\frac{\partial}{\partial y_n}|\tilde{x}-y| = \frac{-(\tilde{x}_n-y_n)}{|x-y|}$$

$$\frac{\partial G}{\partial y_y} = -c_n(2-n)\times$$

$$\left[|x-y|^{1-n}\left(\frac{-(x_n-y_n)}{|x-y|}\right) - |\tilde{x}-y|^{1-n}\left(\frac{-(\tilde{x}_n-y_n)}{|\tilde{x}-y|}\right)\right]$$

On $\partial\mathbb{R}^n_+$, $y_n=0$ and

$|x-y| = |\tilde{x}-y|$, so

$$\frac{\partial G}{\partial y_n} = -c_n(2-n)\frac{[-x_n+\tilde{x}_n]}{|x-y|^n} = \frac{2c_n(2-n)x_n}{|x-y|^n}$$

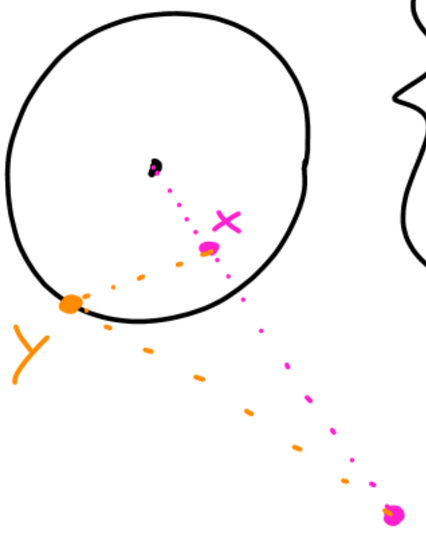
$$u(x) = -\int_{\partial \mathbb{R}^n_+} \frac{\partial G}{\partial \nu_y}(x,y)\, g(y)\, dS(y)$$

$$= -C_n \int_{\mathbb{R}^{n-1}} \frac{x_n}{|x-y|^n} g(y)\, dy_1 \cdots dy_{n-1}$$

<u>Thm</u> If g is $C_0(\mathbb{R}^{n-1})$, then this formula gives a solution to $\begin{cases} -\Delta u = 0 & \text{in } \mathbb{R}^n_+, \\ u = g & \text{on } \partial \mathbb{R}^n_+ \end{cases}$

<u>Pf</u> Thm 14 in Evans

Unit Ball :



$$\begin{cases} -\Delta u = 0 & \text{in } B(0,1) \\ u = g & \text{on } \partial B(0,1) \end{cases}$$

$$G(x,y) = \Phi(x-y) - \Phi(|x|(\tilde{x}-y))$$

$$\tilde{x} = \frac{x}{|x|^2}$$

(Sphere Inversion)

Claim: $\boxed{|x-y| = |x|\,|\tilde{x}-y|}$ if $|y| = 1$.

$$|x-y|^2 = |x|^2|\tilde{x}-y|^2 \ ?$$

$$|x|^2|\tilde{x}-y|^2$$

$$= |x|^2\left|\frac{x}{|x|^2} - y\right|^2$$

$$= |x|^2\left(\left|\frac{x}{|x|^2}\right|^2 + |y|^2 - 2\frac{x\cdot y}{|x|^2}\right)$$

$$= \frac{|x|^2|x|^2}{|x|^4} + |x|^2 - 2x\cdot y$$

$$= 1 + |x|^2 - 2x\cdot y$$

$$|x-y|^2 = |x|^2 + 1 - 2x\cdot y$$

$\Rightarrow$ Claim is true.

$$\Rightarrow \quad c_n |x-y|^{2-n}$$

$$= c_n \left(|x|\,|\tilde{x}-y|\right)^{2-n}$$

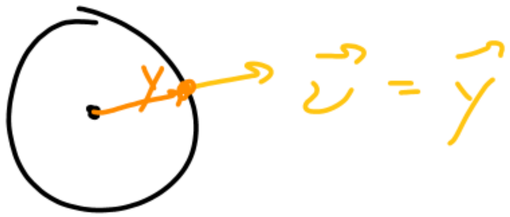
$\Rightarrow$ $G(x,y) = 0$ on $\partial B(0,1)$

$$\Delta_y \phi^x(y) = 0 \text{ in } B(0,1):$$

$$\Delta_y \left(c_n |x|^{2-n} |\tilde{x}-y|^{2-n}\right)$$

$$= c_n |x|^{2-n} \Delta_y |\tilde{x}-y|^{2-n} = 0$$

$$u(x) = -\int_{\partial B(0,1)} \frac{\partial G}{\partial \nu}(x,y)\, g(y)\, dS(y)$$



$$\frac{\partial G}{\partial \nu_y} = D_y G(x,y)\cdot \vec{y}$$

$$= \sum_{i=1}^{n} y_i \frac{\partial}{\partial y_i} G(x,y)$$

$$\frac{\partial}{\partial y_i} G(x,y) = \frac{\partial}{\partial y_i}\left[\Phi(x-y) - \Phi\left(|x|(\tilde{x}-y)\right)\right]$$

$$= C_n \frac{\partial}{\partial y_i}\left[|x-y|^{2-n} - |x|^{2-n}|\tilde{x}-y|^{2-n}\right]$$

$$= C_n(2-n)\left[|x-y|^{1-n}\frac{(x_i-y_i)}{|x-y|} - \frac{|x|^{2-n}(\tilde{x}_i-y_i)}{|\tilde{x}-y|}|\tilde{x}-y|^{1-n}\right]$$

$$= C_n(2-n)\left[\frac{x_i-y_i}{|x-y|^n} - \frac{|x|^{2-n}(\tilde{x}_i-y_i)}{|x-y|^n|x|^{-n}}\right]$$

Using $|\tilde{x}-y| = \frac{|x-y|}{|x|}$.

$$= C_n(2-n)\left(\frac{x_i-y_i - |x|^2(\tilde{x}_i-y_i)}{|x-y|^n}\right)$$

$$D_y G(x,y)\cdot y$$

$$= \sum_{i=1}^{n} y_i \frac{\partial G}{\partial y_i}(x,y)$$

$$= \sum_{i=1}^{n} y_i\, C_n(2-n)\left[\frac{x_i - y_i - |x|^2(\tilde{x}_i - y_i)}{|x-y|^n}\right]$$

$$= C_n(2-n)\left[\frac{\sum x_i y_i - \sum y_i^2 - |x|^2 \sum y_i \tilde{x}_i + |x|^2 \sum y_i^2}{|x-y|^n}\right]$$

$$= C_n(2-n)\frac{\left(x\cdot y - 1 - |x|^2\, \tilde{x}\cdot y + |x|^2\right)}{|x-y|^n}$$

$$\tilde{x} = \frac{x}{|x|^2}$$

$$= C_n \frac{(2-n)(|x|^2-1)}{|x-y|^n}$$

$$x\cdot y - |x|^2\, \tilde{x}\cdot y$$

$$= x\cdot y - x\cdot y = 0$$

$$\frac{\partial G}{\partial \nu_y}(x,y) = \frac{-C_n(1-|x|^2)}{|x-y|^n}$$

Poisson kernel

$$\Rightarrow u(x) = c_n \int_{\partial B(0,1)} \frac{(1-|x|^2)}{|x-y|^n} g(y)\, dS(y)$$

solves $\begin{cases} -\Delta u = 0 & \text{in } B(0,1) \\ u = g & \text{on } \partial B(0,1) \end{cases}$

When $x = 0$,

$$u(0) = c_n \int_{\partial B(0,1)} \frac{1}{|y|^n} g(y)\, dS(y) = c_n \int_{\partial B(0,1)} g(y)\, dS(y)$$

Mean Value Formula

<u>Thm</u> $G(x,y) = G(y,x)$ for $x, y \in U$

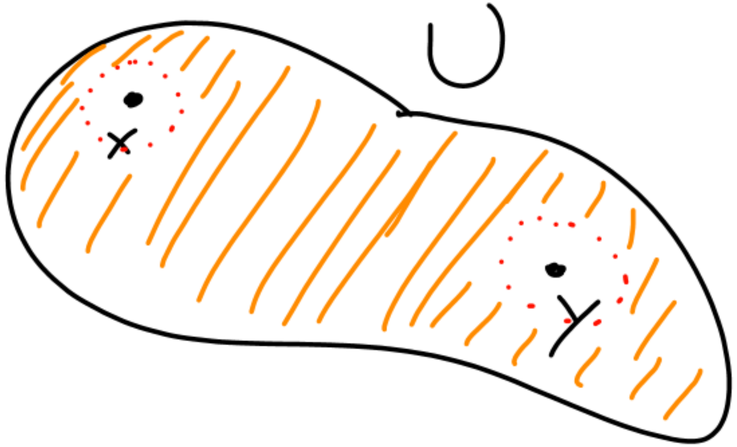
(Consider $\begin{cases} -\Delta u = f & \text{in } U \\ u = 0 & \text{on } \partial U \end{cases}$,

then
$$u(x) = \int_U G(x,y) f(y)\, dy$$

If $f(y) = \delta(y - y_0)$, then $u(x) = G(x, y_0)$

$\Longrightarrow$ Interpret $G(x,y)$ as effect of forcing at y on the solution at x. Theorem $\Rightarrow$ This relationship is symmetric.)

Pf



Green's identity:

$$\int_V (u\Delta v - v\Delta u)\,dz = \int_{\partial V}\left(u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu}\right)dS(z)$$

Choose $V = U \setminus (B(x,\varepsilon)\cup B(y,\varepsilon))$,

$$u = G(x,z)$$

$$v = G(y,z)$$

$$\int_{U\setminus(B(x,\varepsilon)\cup B(y,\varepsilon))}\left(G(x,z)\,\cancelto{0}{\Delta_z G(y,z)} - G(y,z)\,\cancelto{0}{\Delta_z G(x,z)}\right)dz = \int_{\partial(U\setminus B(x,\varepsilon)\cup B(y,\varepsilon))}\left[G(x,z)\frac{\partial G}{\partial \nu_z}(y,z) - G(y,z)\frac{\partial G}{\partial \nu_z}(x,z)\right]dS_z$$